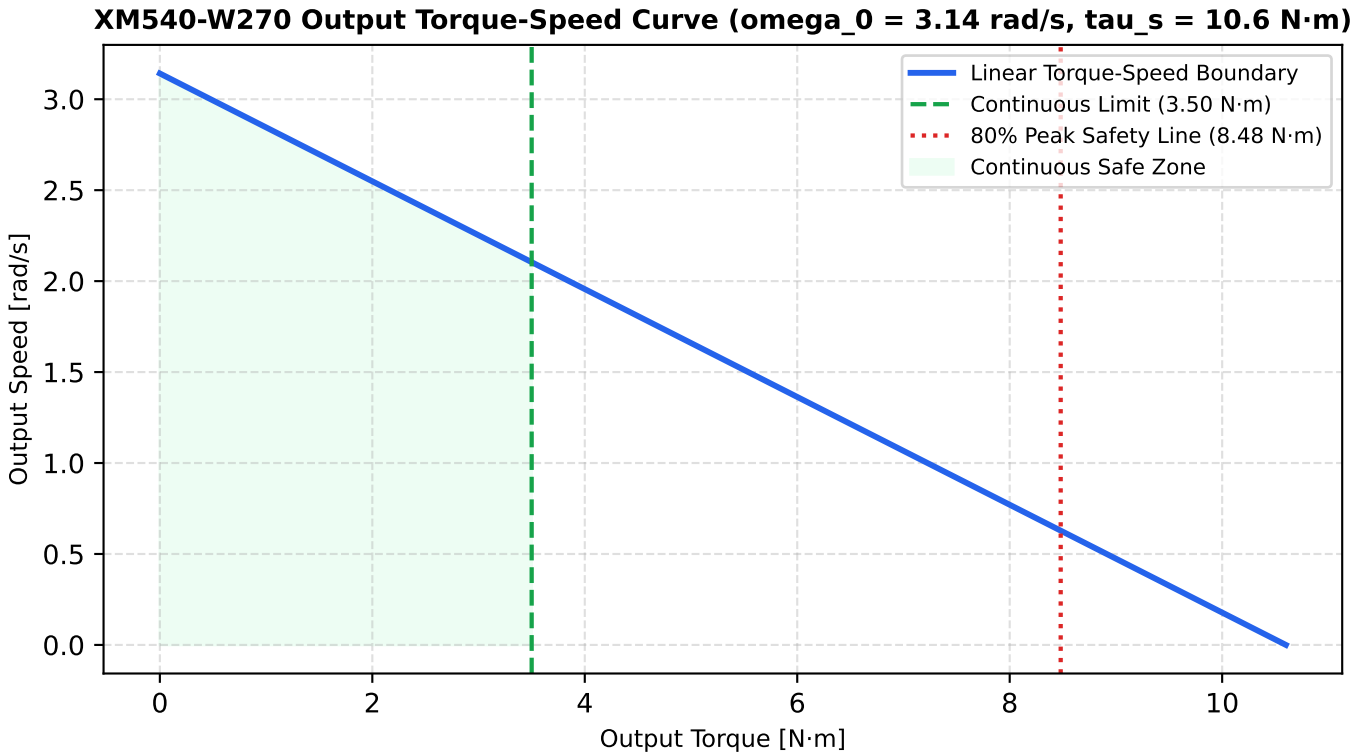
Part Number: XM540-W270

1. Electrical & Mechanical Specifications (Standard Reference)

Parameter	Symbol	Specification Value	Test Condition / Notes
Part Number	ID	XM540-W270	DYNAMIXEL-X
Manufacturer	Vendor	ROBOTIS	Coreless DC Motor
Rated Input Voltage	V_rated	12.0 V	Nominal DC Bus
Stall / Peak Torque	tau_stall	10.6 N.m (at 12.0V, 4.4A)	At 12.0 V
Continuous Rated Torque	tau_cont	3.50 N.m continuous	25 degC ambient, continuous duty
No-Load Speed	omega_0	30 rpm (at 12.0V)	Unloaded output shaft
Actuator Mass (Weight)	m_act	165.0 g	Includes housing & internal gearhead
Internal Reduction	i_int	272.5 : 1	Integrated gear stage
Envelope Dimensions	W x H x D	33.5 x 58.5 x 44.0 mm	Outer housing envelope



XM540-W270 — Mechanical Drawing & Operating Guidelines

2. Mechanical Mounting & Frame Integration
   - Housing Envelope: 33.5 mm x 58.5 mm x 44.0 mm
   - Total Verified Mass: 165.0 g (0.1650 kg)
   - Mounting Pattern: M2.5 / M3 ISO radial & horn flange threads

3. Thermal & Duty-Cycle Notes
   - Continuous operation must remain at or below the continuous rated torque.
   - If continuous torque is not explicitly tabulated on Page 1, apply a 0.33x stall torque
   - Peak torque operation (>80% of available speed-dependent torque) risks step loss or the

4. Citation Verification Hash
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